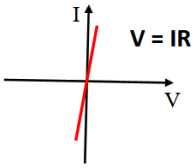
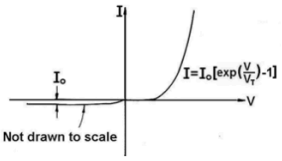
Solid State Devices EE3001 Endsem Syllabus

Metal-Semiconductor junctions

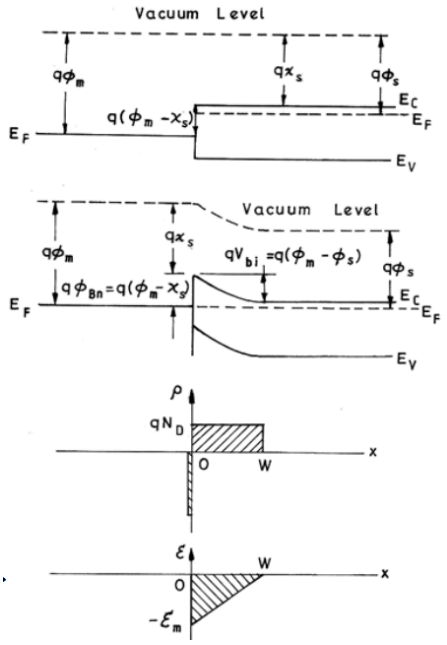


Ohmic



Schottky (rectifying)

Metal/n-type semiconductor ($\phi_m > \phi_s$) junction at thermal eq



Definitions:
$\phi_{Bn} \Rightarrow$ Barrier height (or barrier potential) at heterojunction
$V_{bi} \Rightarrow$ Built-in potential
$W \Rightarrow$ Depletion layer width

$$\phi_{Bn} = \phi_m - \chi_s$$

$$\begin{aligned} V_{bi} &= \phi_m - \phi_s = \phi_{Bn} - (\phi_s - \chi_s) \\ &= \phi_{Bn} - V_T \ln\left(\frac{N_C}{N_D}\right) \end{aligned}$$

$$W = \sqrt{\frac{2\varepsilon V_{bi}}{qN_D}}$$

$$E_m = \frac{qN_D W}{\varepsilon} = \frac{2V_{bi}}{W}$$

Similar to p^+n junction diode

Current flow under bias

Thermal equilibrium

Equal flow of electrons in both directions

$$n_{s0} = N_D \exp\left[-\frac{V_{bi}}{V_T}\right]$$

$$|J_{m\to s}| = |J_{s\to m}| = A^{**}T^2 \exp\left[-\frac{\phi_{Bn}}{V_T}\right] = J_s$$

$$J = |J_{s\to m}| - |J_{m\to s}| = 0$$

Forward bias

Flow of electrons:

- metal to semiconductor unchanged
- semiconductor to metal increases exponentially

$$n_s = N_D \exp\left[-\frac{(V_{bi} - V_F)}{V_T}\right] = n_{s0} \exp\left[\frac{V_F}{V_T}\right]$$

$$|J_{m\to s}| = J_s, \quad |J_{s\to m}| = J_s \exp\left[\frac{V_F}{V_T}\right]$$

$$J = |J_{s\to m}| - |J_{m\to s}| = J_s \left[\exp\left(\frac{V_F}{V_T}\right) - 1\right]$$

Reverse bias

Flow of electrons:

- metal to semiconductor unchanged
- semiconductor to metal reduces

$$n_s = N_D \exp\left[-\frac{(V_{bi} + V_R)}{V_T}\right] = n_{s0} \exp\left[-\frac{V_R}{V_T}\right]$$

$$|J_{m\to s}| = J_s, \quad |J_{s\to m}| = J_s \exp\left[-\frac{V_R}{V_T}\right]$$

$$J = |J_{s\to m}| - |J_{m\to s}| = J_s \left[\exp\left(-\frac{V_R}{V_T}\right) - 1\right]$$

Semiconductor	Work functions	Type
n-type ($< 10^{17}/\text{cm}^3$)	$\phi_m > \phi_s$	Schottky
n-type ($< 10^{17}/\text{cm}^3$)	$\phi_m < \phi_s$	Ohmic
p-type ($< 10^{17}/\text{cm}^3$)	$\phi_m > \phi_s$	Ohmic
p-type ($< 10^{17}/\text{cm}^3$)	$\phi_m < \phi_s$	Schottky
n-type/p-type ($> 10^{19}/\text{cm}^3$)	Does not matter	Ohmic

Bipolar junction Transistors

The Bipolar Junction Transistor (BJT) is a three-terminal solid-state device consisting of an **emitter, a base, and a collector**, structured as either p-n-p or n-p-n.

- For transistor action to occur, the middle p-region (base) must be very narrow—specifically, its width must be less than the diffusion length of electrons.
- When the emitter-base (E-B) junction is forward-biased, it injects electrons into the base, which are then swept across the depletion region of the reverse-biased collector-base (C-B) junction into the collector.

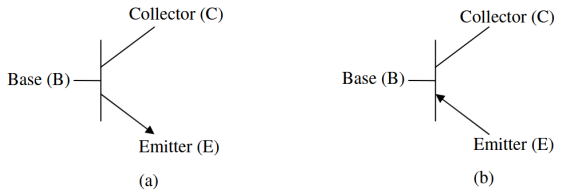


Fig.6.1: Circuit symbol of (a) npn and (b) pnp transistor

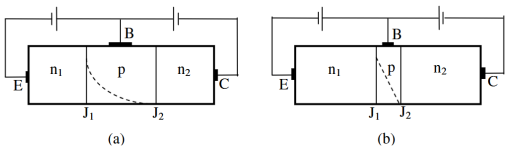
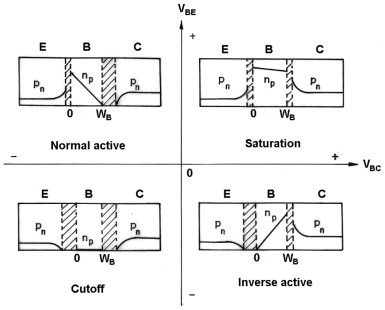


Fig.6.2: n-p-n structures where (a) the width of the p-region is much greater than the diffusion length of electrons and (b) the width of the p-region is much less than the diffusion length of electrons. The electron concentration profile in the p-region (base) is also shown.

(a) Normal active mode – E-B junction is forward biased and C-B junction is reverse biased
(b) Saturation mode – Both E-B and C-B junctions are forward biased,
(c) Inverse active mode – E-B junction is reverse biased and C-B junction is forward biased
(d) Cutoff mode – Both E-B and C-B junctions are reverse biased.



$$I_E = I_B + I_C$$

The sum of the majority and minority carrier components at the contact equals the sum of the minority carrier current components at the junction.

- **Emitter Current** (I_E) : Consists of hole current (I_{pE}) and electron current (I_{nE}) at the emitter-base (E-B) ju
- **Collector Current** (I_C) : Consists of hole current (I_{pC}) and electron current (I_{nC}) at the collector-base (C-B)
- **Base Current** (I_B) : Comprises $I_{pE}, I_{pC},$ and the base recombination current $I_{BR} = -(I_{nE} - I_{nC})$.

The currents at each terminal are composed of specific electron and hole flows across the junctions.
- I_E, I_B, I_C: Total current at the Emitter, Base, and Collector terminals.
- I_{nE}: Electron current flowing at the E-B junction.
- I_{pE}: Hole current at the E-B junction (holes injected from base into emitter during forward bias).
- I_{nC}: Electron current flowing at the C-B junction.
- I_{pC}: Hole current flowing at the C-B junction.
- I_{BR}: Base recombination current (accounts for electrons "lost" to recombination in the base).

Transistor Current Gains (α and β)

Theory: High gain requires high **emitter efficiency** ($\gamma \approx 1$) and a high **base transport factor** ($\alpha_T \approx 1$).
Key Formulas
- **Common-Base Gain:** $\alpha = \frac{I_{nC}}{I_E} = \gamma \alpha_T$
- **Common-Emitter Gain:** $\beta = \frac{\alpha}{1-\alpha}$ or $\alpha = \frac{\beta}{1+\beta}$
- **Collector Output (CB):** $I_C = \alpha I_E + I_{CBO}$
- **Collector Output (CE):** $I_C = \beta I_B + (\beta + 1) I_{CBO}$
- **Emitter Efficiency (γ):** $\gamma \approx \left[1 + \frac{N_{DE} W_E}{N_A W_B}\right]^{-1}$
- **Base Transport Factor (α_T):** $\alpha_T \approx \left[1 + \frac{W_B^2}{2L_n^2}\right]^{-1}$

• **Exact γ (Long Emitter):** $\gamma = \left[1 + \frac{N_{DE} D_{nB} L_{pE}}{N_A D_{pE} W_B} \coth\left(\frac{W_E}{L_{pE}}\right)\right]^{-1}$

Terminology

• I_{CBO}: Open-circuit leakage current (C-B junction leakage when $I_E = 0$).
• N_A, N_{DE}: Doping concentrations in the Base (Acceptors) and Emitter (Donors).
• W_B, W_E: Physical widths of the neutral Base and Emitter regions.
• L_n, L_{pE}: Minority carrier diffusion lengths.
• τ_n: Minority carrier lifetime in the base; τ_{tB}: Base transit time.

The Ebers-Moll Model

Simulates BJT in all modes as two back-to-back diodes interacting via coupled current sources.

Key Formulas

• **Forward/Reverse Diode Current:**
$I_F = I_{ES}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \ | \ I_R = I_{CS}\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right]$
• **Terminal Currents:**
$I_E = I_F - \alpha_R I_R$
$I_C = \alpha_F I_F - I_R$
$I_B = (1 - \alpha_F) I_F + (1 - \alpha_R) I_R$
• **Floating Collector ($I_C = 0$):**
$V_{CE} = V_T \ln\left(\frac{1}{\alpha_R}\right)$
$V_{BC} \approx V_{BE} + V_T \ln(\alpha_R)$

- *IF*,*IR*: The forward and reverse currents flowing through the back-to-back diodes representing the Emitter-Base (E-B) and Collector-Base (C-B) junctions.
- *IES*,*ICS*: The reverse saturation currents of the specific E-B and C-B junction diodes, respectively.
- *VBE*,*VBC*: The applied voltages across the emitter-base and collector-base junctions.
- *VT*: The thermal voltage (used in standard diode exponential equations).
- *αF*,*αR*: The forward and reverse common-base current gains.
- *βF*,*βR*: The forward and reverse common-emitter current gains. *βF* applies when *VBC*=0, and *βR* applies when *VBE*=0.
- *IS*: The saturation current parameter used to relate terminal currents to voltages across all combinations of *VBE* and *VBC*.
- *q*: Elementary charge.
- *A*: The cross-sectional area of the transistor.
- *Dn*: The minority carrier diffusion constant for electrons.
- *ni*: The intrinsic carrier concentration.
- *ICT*: The current supplied by the single coupled current source in the simplified three-parameter Ebers-Moll circuit.
- *ICC*,*IEC*: Components of the forward and reverse diode currents used in the unified Ebers-Moll equivalent circuit.

Non-Ideal Effects (Early Effect & Breakdown)

Reverse bias on the C-B junction widens the depletion layer, shrinking W_B (Early Effect). Excessive bias leads to **Punch-Through** or **Avalanche** breakdown.

Key Formulas

• **Early Voltage:** $V_A = W_B(0)\left[\frac{\partial V_{CB}}{\partial W_B}\right]^{-1}$
• **Punch-Through Voltage:** $V_{PT} = \frac{qN_A W_B^2}{2\epsilon_s}$

- **Avalanche Factor (M):** $M = \left[1 - \left(\frac{V_{CB}}{V_{CBO}}\right)^m\right]^{-1}$
- **CE Breakdown Voltage:** $V_{CEO} \approx \frac{V_{CBO}}{(1+\beta)^{1/m}}$

- *VA*: The Early Voltage. It is the parameter where extrapolated common-emitter characteristics (in the active region) intersect on the negative voltage axis.
- *WB*(0): The neutral base width when the voltage across the C-B junction (*VBC*) is zero.
- *gcb*,*gce*: The output conductances in the common-base and common-emitter modes, respectively. They measure how much the output current changes as the output voltage changes.
- *VPT*: The base punch-through voltage. This is the voltage at which the depletion region extends across the entire base, reducing the neutral base width to zero.
- *ϵs*: The permittivity of the semiconductor material.
- *M*: The collector avalanche multiplication factor, which accounts for the multiplication of current flowing out of the collector-base depletion region under large reverse bias.
- *m*: An empirical constant dependent on the junction material and grading.
- *VCBO*: The maximum breakdown voltage that can be applied across the C-B junction with the emitter open.
- *VCEO*: The maximum breakdown voltage that can be applied across the collector-emitter junction with the base open. It is considerably smaller than *VCBO*.

Capacitances

Charge storage limits switching speed. **Diffusion capacitance** (C_D) is due to minority carriers; **Junction capacitance** (C_J) is due to the depletion region.

Key Formulas

- **Diffusion Capacitance:** $C_{DE} = \frac{\tau_F I_{CC}}{V_T}$
- **Forward Transit Time:** $\tau_F = \frac{W_B^2}{2D_{nB}} + \frac{W_E W_B N_A}{2D_{pE} N_{DE}}$
- **Junction Capacitance:** $C_J = C_{J0}\left(1 - \frac{V}{V_{bi}}\right)^{-m}$

- *CDE*,*CDC*: The diffusion capacitances for the E-B and C-B junctions. They arise from stored mobile minority charges in the neutral regions.
- *τF*,*τR*: The forward and reverse transit times of the transistor, which are constant for a particular device structure.
- *CJE*,*CJC*: The junction (depletion) capacitances at the E-B and C-B junctions, caused by fixed charges in the depletion regions.
- *CJE*0,*CJC*0: The zero-bias values of the E-B and C-B junction capacitances (capacitance when *VBE*=0 and *VBC*=0).
- *VbiE*,*VbiC*: The built-in potentials of the E-B and C-B junctions.
- *mE*,*mC*: The grading factors of the E-B and C-B junctions, dependent on the nature of the doping transition.

Switching Characteristics

BJT switching speed is limited by the injection/removal of minority carriers. The **Storage Delay** (t_{sd}) is the longest delay.

Key Formulas

- **Base Charge Equation:** $\frac{dQ_{nB}}{dt} = I_{FB} - \frac{Q_{nB}}{\tau_n}$
- **Turn-on Time:** $t_{on} = \tau_n \ln\left(1 - \frac{I_{Csat}}{\beta I_{FB}}\right)^{-1}$
- **Storage Delay Time:** $t_{sd} = \tau_n \ln\left(\frac{I_{FB} + I_{RB}}{\frac{I_{Csat}}{\beta} + I_{RB}}\right)$
- **Total Delay:** $t_{delay} = t_{on} + t_{sd} + t_{dis}$

Circuit Definitions

• $I_{Csat} = \frac{V_{CC} - V_{CEsat}}{R_C}$
• $I_{FB} \approx \frac{V_{in(High)}}{R_B} \mid I_{RB} \approx \frac{-V_{in(Low)}}{R_B}$

- *QnB*: The excess minority carrier charge physically stored in the neutral base.
- *IFB*: The forward base current driving the transistor into the ON state.
- *IRB*: The reverse base current flowing out of the base, driving the transistor into the OFF state.
- *ICsat*: The saturation collector current limit imposed by the external circuit (approximately *VCC*/*RL*).
- *ton*: The turn-on time, which is the delay required for the collector current to reach its saturation limit.
- *tsd*: The storage delay time. It is the duration the transistor remains in saturation *after* the input voltage drops, waiting for excess stored charge to be removed.
- *tdis*: The discharge time required for the transistor to transition from the edge of saturation down to cut-off.
- *toff*: The time from when the input voltage switches until the excess stored base charge reaches exactly zero.
- *VCC*: The external circuit supply voltage.
- *VCEsat*: The minimal voltage across the collector-emitter terminals when the BJT is in deep saturation (typically around 0.2V).
- *RL*,*RB*,*RC*: The Load, Base, and Collector resistors used in switching or inverter circuits.
- *τn*: The minority carrier (electron) lifetime in the base. All switching delays are directly proportional to this value.

MOSFETS

A Metal-Oxide-Semiconductor structure that behaves as a voltage-variable capacitor. Its operating state is dictated by the Gate-to-Bulk voltage (V_{GB}).

Operating Modes

• **Flat-Band:** No net charge in the semiconductor; energy bands are flat.
• **Accumulation:** Negative V_{GB} (for p-type) attracts majority holes to the surface.
• **Depletion:** Small positive V_{GB} repels holes, leaving fixed negative acceptor ions.
• **Inversion:** High positive V_{GB} attracts minority electrons to form a channel.
• **Strong Inversion:** Occurs when surface potential $\psi_s = 2\phi_B$.

Key Equations

$$\text{Fermi Potential:} \quad \phi_B = V_T \ln\left(\frac{N_A}{n_i}\right)$$

$$\text{Max Depletion Width:} \quad W_{max} = \sqrt{\frac{2\epsilon_s(2\phi_B)}{qN_A}}$$

$$\text{Flat-Band Voltage:} \quad V_{FB} = \phi_{ms} - \frac{Q_f + Q_m + Q_{ot}}{C_{ox}}$$

$$\text{Threshold Voltage (Ideal):} \quad (V_{th})_{ideal} = 2\phi_B + \frac{\sqrt{2qN_A\epsilon_s(2\phi_B)}}{C_{ox}}$$

Capacitance-Voltage (C-V) Characteristics

The device acts as two capacitors in series: the oxide capacitance (C_{ox}) and the semiconductor depletion capacitance (C_s).

• **Accumulation:** $C \approx C_{ox}$.
• **Depletion:** C drops as W increases.
• **Inversion (HF):** Capacitance stays at C_{min} because minority carriers cannot follow the high-frequency signal.

Key Equations

$$\text{Total Capacitance:} \quad \frac{1}{C} = \frac{1}{C_{ox}} + \frac{1}{C_s} = \frac{1}{C_{ox}} + \frac{W}{\epsilon_s}$$

$$\text{Minimum Capacitance (HF):} \quad \frac{1}{C_{min}} = \frac{1}{C_{ox}} + \frac{W_{max}}{\epsilon_s}$$

$$\text{Doping Extraction:} \quad \frac{1}{(C')^2} = \frac{1}{(C'_{ox})^2} + \frac{2(V_{GB} - V_{FB})}{A^2 q N_A \epsilon_s}$$

MOSFET & Body Effect

Applying a substrate bias (V_{BS}) changes the depletion width and shifts the threshold voltage.

Key Equations

$$\text{Body-Effect Parameter:} \quad \gamma = \frac{\sqrt{2\epsilon_s q N_A}}{C_{ox}}$$

$$\text{Threshold Voltage shift:} \quad \Delta V_{th} = \gamma \left(\sqrt{2\phi_B - V_{BS}} - \sqrt{2\phi_B} \right)$$

$$\text{General Threshold:} \quad V_{th} = V_{FB} + 2\phi_B + \gamma \sqrt{2\phi_B - V_{BS}}$$

MOSFET I-V Models

Models for drift current once $V_{GS} > V_{th}$.

SPICE Level 1 (Square Law)

$$\text{Linear Region:} \quad I_D = \frac{Z}{L} \mu_n C_{ox} \left[(V_{GS} - V_{th}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

$$\text{Saturation Voltage:} \quad V_{Dsat} = V_{GS} - V_{th}$$

$$\text{Saturation Current:} \quad I_{Dsat} = \frac{Z}{2L} \mu_n C_{ox} (V_{GS} - V_{th})^2$$

SPICE Level 3 (Semi-Empirical)

$$\text{Correction Factor:} \quad \alpha = 1 + \frac{\gamma}{2\sqrt{2\phi_B - V_{BS}}}$$

$$\text{Saturation Current:} \quad I_{Dsat} = \frac{Z}{2\alpha L} \mu_n C_{ox} (V_{GS} - V_{th})^2$$

Terminologies

Symbol	Description
I_E, I_B, I_C	Total terminal currents (Emitter, Base, Collector)
ϕ_B	Fermi Potential: $V_T \ln(N_A/n_i)$
V_{FB}	Flat-band Voltage
C_{ox}	Oxide capacitance per unit area
γ	Body-effect parameter
μ_n	Electron mobility in the channel
W_{max}	Maximum depletion width at strong inversion
ϕ_{ms}	Metal-semiconductor work function difference
α, β	Common-base and Common-emitter current gains